

Employment Database – MySQL

Assignment 2

This document summarizes the supplied employment_db SQL work and the query requirements from the uploaded MySQL Assignment 2 – Querying Data document.

Assignment Basis

The assignment continues from the previous database/table creation work, asks the student to populate the tables using the provided DML, and then execute the required queries.

Database Structure

- departments / Departments_Info – department master data
- Location / Locations – location master data
- Employee – employee information with department and location foreign keys

SQL Sequence Note

The supplied SQL alters and renames tables/columns, but later INSERT and SELECT statements still reference some old names such as departments, age, and hire_date. Therefore, the SQL should be executed in a consistent sequence or those later references should be updated to the final names before running the complete script.

DISTINCT

Retrieve distinct salaries.

```
SELECT DISTINCT salary FROM Employee;
```

Alias (AS)

Alias age as Employee_Age and salary as Employee_Salary.

```
SELECT age AS Employee_Age FROM Employee;  
SELECT salary AS Employee_Salary FROM Employee;
```

WHERE Clause

Retrieve salaries greater than 50000 and employees hired before 2016-01-01.

```
SELECT * FROM Employee WHERE salary > 50000;  
SELECT * FROM Employee WHERE hire_date < '2016-01-01';
```

Missing Designation

Find the missing designation and update employee 5004 to Data Scientist.

```
SELECT * FROM Employee WHERE designation IS NULL;
UPDATE Employee SET designation = 'Data Scientist' WHERE employee_id = '5004';
```

ORDER BY

Sort by department ID ascending and salary descending.

```
SELECT * FROM Employee ORDER BY department_id ASC;
SELECT * FROM Employee ORDER BY salary DESC;
```

LIMIT

Display the first five employees hired in 2018.

```
SELECT * FROM Employee WHERE hire_date >= '2018-01-01' AND hire_date < '2019-01-01' ORDER BY hire_date LIMIT 5;
```

Aggregate Functions

Calculate Finance total salary and minimum age.

```
SELECT SUM(salary) AS total_salary FROM Employee WHERE department_id = 7;
SELECT MIN(age) AS age_employee FROM Employee;
```

GROUP BY

Maximum salary for each location and average salary for Analyst designations.

```
SELECT location_id, MAX(salary) AS max_sal FROM Employee GROUP BY location_id;
SELECT designation, AVG(salary) AS avg_sal_designation FROM Employee WHERE designation LIKE '%Analyst%' GROUP BY designation;
```

HAVING

Find departments with fewer than three employees and female locations with average age below 30.

```
SELECT department_id, COUNT(*) AS department_count FROM Employee GROUP BY department_id HAVING COUNT(*) < 3;
SELECT location_id, AVG(age) AS average_age FROM Employee WHERE gender = 'F' GROUP BY location_id HAVING AVG(age) < 30;
```

INNER JOIN

List employee name, designation, and department name.

```
SELECT e.employee_name, e.designation, d.department_name FROM Employee AS e INNER JOIN departments AS d ON e.department_id = d.department_id;
```

LEFT JOIN

List all departments with employee counts, including empty departments.

```
SELECT d.department_name, COUNT(e.employee_id) AS employee_count FROM departments AS d LEFT JOIN Employee AS e ON e.department_id = d.department_id GROUP BY d.department_id, d.department_name;
```

RIGHT JOIN

Display all locations and assigned employees, retaining locations without employees.

```
SELECT l.location_name, e.employee_name FROM Employee AS e RIGHT JOIN Locations  
AS l ON e.location_id = l.location_id;
```

GitHub Repository Structure

Recommended:

- README.md
- employment_db.sql
- MySQL Assignment 2 – Querying Data.pdf

Source

Based on the uploaded MySQL Assignment 2 – Querying Data document. The source specifies the DISTINCT, alias, WHERE, sorting, LIMIT, aggregate, GROUP BY, HAVING, and JOIN tasks.

The screenshot displays a database query interface. At the top, there's a toolbar with options like 'Result Grid', 'Filter Rows', 'Edit', 'Export/Import', and 'Wrap Cell Content'. Below this is a table with 10 columns: employee_id, employee_name, Gender, Age, hire_date, designation, salary, department_id, and location_id. The table contains 7 rows of data. Below the table, there are tabs labeled 'EMPLOYEE 6' and 'EMPLOYEE 7'. Below the tabs, a SQL query is shown:

```
112 SELECT location_id, MAX(salary) AS max_sal  
113 FROM employee
```

Below the query, there's another toolbar with 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below this is a small table with 2 columns: total_salary and 170000.00.

employee_id	employee_name	Gender	Age	hire_date	designation	salary	department_id	location_id
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	4	4
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3	4
5012	Myra Verma	F	26	2016-11-05	Operations Manager	95000.00	13	2
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	95000.00	5	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5030	Sara Iyer	F	28	2019-11-20	Project Manager	90000.00	5	1

total_salary
170000.00

```

109 --Minimum---
110 Select Min(age) as age_employee From employee;
111 --maximum salary for each location---
112 SELECT Location_id, MAX(salary) AS max_sal
113 FROM employee
114 GROUP BY Location_id;
115 --Average salary for each designation containing the word 'Analyst'---
116 Select designation, avg(salary) as avg_sal_designation

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

age_employee
25

```

112 SELECT Location_id, MAX(salary) AS max_sal
113 FROM employee
114 GROUP BY Location_id;
115 --Average salary for each designation containing the wo
116 Select designation, avg(salary) as avg_sal_designation
117 From employee
118 Where designation like '%analyst%'
119 Group by designation;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Location_id	max_sal
▶	1	95000.00
	2	95000.00
	3	90000.00
	4	100000.00

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Content:	
	employee_name	designation	department_name
▶	Anvi Chaudhary	Business Development Executive	Business Development
	Kiaan Desai	Sales Executive	Business Development
	Ananya Paul	Content Writer	Content Creation
	Anaya Kapoor	Event Coordinator	Content Creation
	Aaradhya Iyer	Customer Support Executive	Customer Support
	Arnav Rao	Customer Success Manager	Customer Support
	Vihaan Mohan	Supply Chain Analyst	Customer Support

Result 13 x

```

127      from employee as e
128      Inner join departments as d on e.department_id = d.department_id;
129 ✖ ----List all departments along with the total number of employees in each
130      Select e.employee_name, d.department_name
131      From departments as d
132      left join employee as e on e.department_id = d.department_id;
133 •      Select d.department_name,
134      count(e.employee_id) as employee_count

```

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Content:	
	employee_name	department_name	
▶	Anvi Chaudhary	Business Development	
	Kiaan Desai	Business Development	
	Ananya Paul	Content Creation	
	Anaya Kapoor	Content Creation	
	Aaradhya Iyer	Customer Support	
	Arnav Rao	Customer Support	Customer Support
	Vihaan Mohan	Customer Support	

Result 14 x

Output

```

137      GROUP BY d.department_id, d.department_name;
138 •   Select l.Location_name, e.employee_name
139      From locations as l
140      Right join employee as e on e. location_id =
141 •   delete from departments

```

Result Grid   Filter Rows: Export:  Write

	Location_name	employee_name
▶	Pune	Vihaan Singh
	Chennai	Reyansh Singh
	Bangalore	Aradhya Iyer
	Hyderabad	Kiara Malhotra
	Chennai	Anvi Chaudhary
	Bangalore	Dhruv Shetty
	Hyderabad	Anushka Singh

Result 15 x

Output